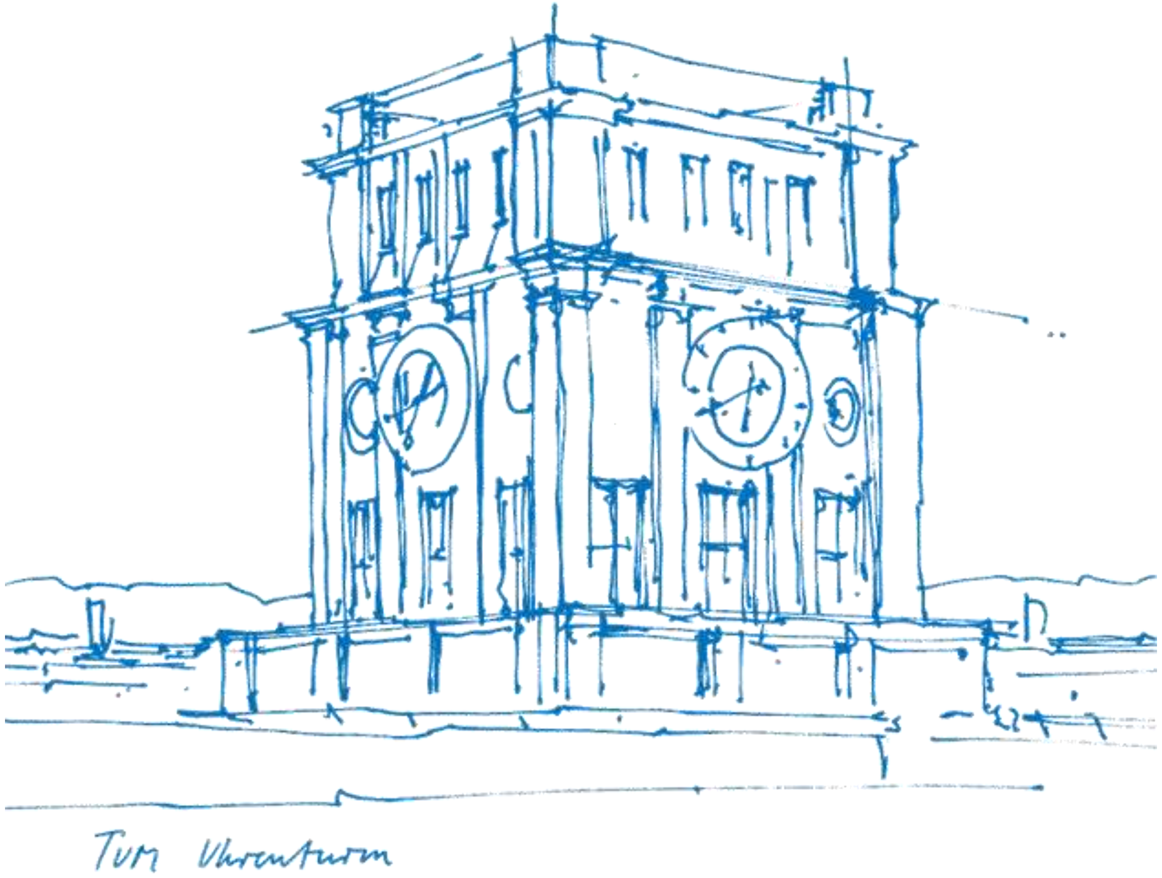
Discrete Probability Theory (IN0018)
School of Computation, Information and Technology
Technical University of Munich

Tutorial 04

Conditional Probability

Xujie Zhao

25. May / 26. May 2026



About Tutorial

- **Tutorial:** In-Person, 1.5h by two time periods per week - 12:15-13:45 Mon, 00.08.053 (A31)
  - 14:15-15:45 Tue, 00.08.053 (B42)

- The slides are not guaranteed to be accurate! (Written by myself)

- **Materials of this tutorial:** Please scan the QR-Code ☞

- **Contact:** xujie.zhao@tum.de

➢ PART I. Recap of the Lecture

➢ PART II. Exercises from the Exercise Sheet

➢ PART III. Additional Exercises (with solution)

I. Recap

- **(Review)**
  - Inclusion-exclusion principle
  - Cumulative distribution function (cdf)
  - Probability density function (pdf)

- **Conditional Probability**
  - Definition
  - Trace of a probability space
  - Total probability
  - Bayes formula

II. Exercise Sheet

Exercise 1. We consider a fair tetrahedron whose faces are labeled with the numbers $\{1,2,3,4\}$. Assume the tetrahedron is rolled $k \geq 1$ times. Let E be the event that the same number appears in each roll. Furthermore, let F be the event that the product of the k rolled numbers is even.

(i) Compute the probability of E.

(ii) Compute the probability of F.

(iii) Compute the probability of $E \cup F$.

II. Exercise Sheet

Exercise 2. Let $(\Omega, \mathcal{A}, P)$ be a probability space, $B \in \mathcal{A}$ with $P(B) > 0$ and I a countable index set. Show the following statements:

(i) The map

$$\widetilde{P} : \mathcal{A} \to [0,1]\,, \quad A \mapsto \widetilde{P}(A) := P(A \mid B)$$

is a probability measure on $(\Omega, \mathcal{A})$. The probability space $(B, \mathcal{A}|_B, \widetilde{P}|_B)$ is called *trace of $(\Omega, \mathcal{A}, P)$ on B*.

(ii) Let $\{B_i \mid i \in I\} \subseteq \mathcal{A}$ be a measurable *partition* of B, i.e., $\bigcup_{i\in I} B_i = B$ and $B_i \cap B_j = \emptyset$ for all $i, j \in I$ with $i \neq j$. For all $A \in \mathcal{A}$ we then have the *law of total probability*

$$P(A \cap B) = \sum_{i\in I} P(A \mid B_i) P(B_i)\,.$$

(iii) For each $A \in \mathcal{A}$ with $P(A) > 0$ and every measurable partition $\{E_i \mid i \in I\}$ of Ω with $P(E_i) > 0$ for all $i \in I$, we have the *Bayes rule*

$$P(E_i \mid A) = \frac{P(A \mid E_i) P(E_i)}{\sum_{j\in I} P(A \mid E_j) P(E_j)} \quad \text{for all } i \in I\,.$$

II. Exercise Sheet

Exercise 3. For each of the following settings provide all modeling details and rigorous computations for your results.

(i) You play Russian roulette with a six shooter revolver and there are precisely two bullets in neighboring chambers. Your opponent goes first, spins the barrel, pulls the trigger and the gun does not fire. Now it is your turn. Should you spin the barrel again or pull the trigger right away? What is the probability of survival in each case?

(ii) I have two cards in a closed opaque box, one is red on both sides and the other is red on one and blue on the other side. I pull out one card and place it on the table and a red side faces up. What's the probability that the other side is also red?

Exercise 4. We uniformly choose a real number from $[0,1]$, square it, and represent the result by a real-valued random variable X.

(i) What is the cumulative distribution function of X?

(ii) What is the probability density function of X?

(iii) What is the probability that $X > \frac{1}{4}$. Try to find an answer in your head intuitively first.

III. Additional Exersices

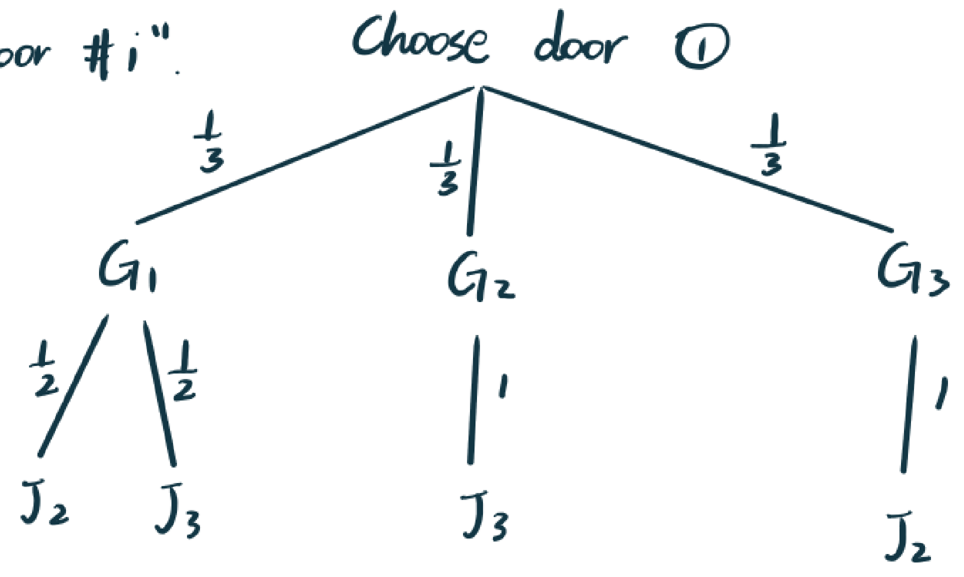
Monty Hall Problem (https://en.wikipedia.org/wiki/Monty_Hall_problem)

Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat. He then says to you, "Do you want to pick door No. 2?" Is it to your advantage to switch your choice?

Define G_i as "car behind door #i", J_i as "host opens door #i".

$$P(G_1 \mid J_3) = \frac{P(G_1 \cap J_3)}{P(J_3)} = \frac{\frac{1}{3} \times \frac{1}{2}}{\frac{1}{3} + \frac{1}{3} \times \frac{1}{2}} = \frac{\frac{1}{6}}{\frac{1}{2}} = \frac{1}{3}.$$

=> Swithing to another door is better!



Materials